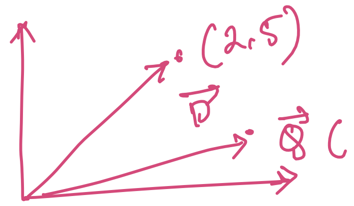
Support Vector Machine

Coordinate Geometry



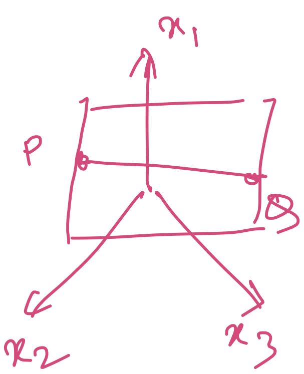
magnitude $= \sqrt{2^2+5^2} = \sqrt{29}$

$\vec{PQ} = \vec{Q} - \vec{P} = (5,5)$

$\vec{Q}\,(7,10)$

$\vec{P}\,(2,5)$

Angle $\tan\theta = y/x \Rightarrow \theta = \tan^{-1}(y/x)$



$f(x) = w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0$

$w^Tx + w_0 = 0$

$w = \{w_1, w_2 \dots w_3\}$

$x = \{x_1, x_2 \dots x_3\}$

Normal vector of $f(x)$

$\vec{w} = [w_1, w_2, w_3] \Rightarrow \nabla f(x)$

show that $\vec{PQ} \cdot \vec{w} = 0$

$\vec{P} = w_1p_1 + w_2p_2 + w_3p_3 + w_0 \Rightarrow w^TP = -w_0$

$\vec{Q} = w_1q_1 + w_2q_2 + w_3q_3 + w_0$

$w^TQ = -w_0$

$w^TP - w^TQ = 0$

$\Rightarrow w^T(P-Q) = 0$

$= w^T \cdot \vec{QP} = 0$

Distance of a point z from the plane, $w^Tx + w_0 = 0$

$$d = \frac{w^Tz + w_0}{|w_0|} = \gamma = \text{geometric distance}$$

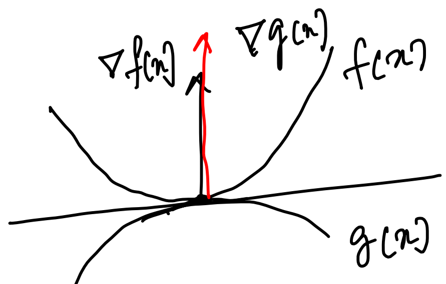
Functional distance $w^Tz + w_0 = \hat{\gamma}$

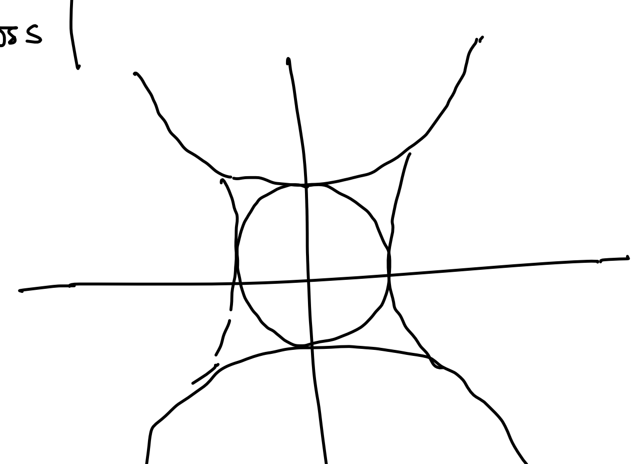
04/09/25 **Lagrange Multiplier**

$\max_{x,y}\ z = x^2y \Leftarrow$

s.t. $x^2 + y^2 = 1$

Regression – linear – Sq loss
– logistic – log loss





$$\boxed{\nabla f(x) = \lambda \nabla g(x)}$$

$\Rightarrow f(x) = x^2 + y^2 - 4$ $\quad g(x) = \frac{x^2}{4} + y^2 - 1$: Ellipse

Circle $\quad$ at $(2,0)$

$\nabla f(x) = (2x, 2y) =$ $\quad \nabla g(x) = (x/2, 2y)$

at $(2,0)$ $\quad (4,0)$ $\quad (1,0)$

Lagrange Eq. $\quad \mathcal{L}(x,\lambda) = f(x) - \lambda g(x)$ $\quad \lambda$: langrange multiplier.

max $f(x)$
s.t $g(x)$
equality

and $\nabla_x \mathcal{L} = 0$

$\nabla_\lambda \mathcal{L} = 0$

$f(x) - \lambda_1 g_1(x) - \lambda_2 g_2(x)$

$\nabla_x \mathcal{L} = 0$

$\nabla_{\lambda_i} \mathcal{L} = 0$

$\max_{h,s} f(h,s) = 200 h^{2/3} s^{1/3}$

s.t.c $\quad 20h + 170s = 20{,}000$

$\mathcal{L}(h,s,\lambda) = 200 h^{2/3} s^{1/3} - \lambda(20h + 170s - 20000)$

$\frac{\partial \mathcal{L}}{\partial h} = 200 \times \frac{2}{3} h^{-1/3} s^{1/3} - 20\lambda = 0$

$\frac{\partial \mathcal{L}}{\partial s} = 200 \times \frac{1}{3} h^{2/3} s^{-2/3} - 170\lambda = 0$

$\frac{\delta \mathcal{L}}{\delta \lambda} = -20h - 170s + 20000 = 0$

$h = 666.66$
$s = 39.12$
$\lambda = 3.59$
$f(\,)_{max} = 51777$

KKT Theory
Karush Kuhn Tucker

max $f(x)$
s.t.c. $h_i(x) \leq 0$

① Convert it to LM
take derivative w.r.t. variable and equate them to 0

② $\boxed{\lambda_i h^i = 0}$ *

③ $h^i \leq 0$

④ $\lambda_i \geq 0$

$\max: -x_1^2 - x_2^2 - x_3^2 + 4x_1 + 6x_2$

s.t. $\quad x_1 + x_2 \leq 2$

$2x_1 + 3x_2 \leq 12$

$x_1, x_2 \geq 0$

Cond^1: $\quad \mathcal{L}(\quad) = -x_1^2 - x_2^2 - x_3^2 + 4x_1 + 6x_2 - \lambda_1(x_1 + x_2 - 2) - \lambda_2(2x_1 + 3x_2 - 12)$

$\frac{\partial \mathcal{L}}{\partial x_1} = -2x_1 + 4 - \lambda_1 - 2\lambda_2 = 0$ — (1a)

$$\frac{\partial L}{\partial x_2} = -2x_2 + 6 - \lambda_1 - 3\lambda_2 = 0 \quad — \text{(1b)}$$

$$\frac{\partial L}{\partial x_3} = -2x_3 = 0 \Rightarrow x_3 = 0$$

Cond^n 2

$$\lambda_1 (x_1 + x_2 - 2) = 0 \quad — \text{(2a)}$$

$$\lambda_2 (2x_1 + 3x_2 - 12) = 0 \quad — \text{(2b)}$$

Cond 3

$$x_1 + x_2 - 2 \le 0 \quad — \text{(3a)}$$

$$2x_1 + 3x_2 - 12 \le 0 \quad — \text{(3b)}$$

Cond 4 : $\lambda_1, \lambda_2 \geq 0$

Case 1: $\lambda_1 = 0$ $\lambda_2 = 0$ (1a) (1b) $x_1 = 2$, $x_2 = 3$

X (3a) $3 \le 0$ X

Case 2 $\lambda_1 \neq 0$, $\lambda_2 \neq 0$; $\left.\begin{array}{l} x_1 + x_2 - 2 = 0 \\ 2x_1 + 3x_2 - 12 = 0 \end{array}\right\}$ $x_1 = 8$ $x_2 = -6$

(1a) (1b) $\lambda_2 = -26$ X

Case 3 $\lambda_1 = 0$, $\lambda_2 \neq 0$ (1a) (1b) $\left.\begin{array}{l} -2x_1 + 4 - 2\lambda_2 = 0 \\ -2x_2 + 6 - 3\lambda_2 = 0 \end{array}\right\}$ $x_1 = \frac{2}{3} x_2$

(2b) $2x_1 + 3x_2 - 12 = 0$

$\Rightarrow x_1 = 2$; $x_2 = 3$ (3a) $5 - 2 \le 0$ X

Case 4 $\lambda_1 \neq 0$ $\lambda_2 = 0$ $x_1 = 1/2$ $x_2 = 3/2$ $\lambda_2 = 0$ $\lambda_1 = 3$

Primal and Dual theorem

$\min_{w} f(w)$

w.r.t. $g_i(w) \le 0$ $i = 1 \dots k$

$h_i(w) = 0$ $i = 1 \dots l$

Generalized Lagrange Eq^n: $\mathcal{L}(w, \alpha, \beta) = f(w) + \sum_{i=1}^{k} \alpha_i g_i(w) + \sum_{i=1}^{l} \beta_i h_i(w)$

Let us define: $\theta_p(w) = \max_{\substack{\alpha, \beta \\ \alpha \geq 0}} \mathcal{L}(w, \alpha, \beta)$

$$= \max_{\substack{\alpha, \beta \\ \alpha \geq 0}} f(w) + \sum \alpha_i g_i(w) + \sum \beta_i h_i(w)$$

Cond^n Violated $g_i(w) > 0$ $\alpha_i \to \infty$ $\theta_p(w) \to \infty$

$h_i(w) \neq 0$ $\beta_i \to +\infty / -\infty$ $\theta_p(w) \to \infty$

Cond^n Satisfied $\theta_p(w) = f(w)$ | $\theta_p(w) = \begin{cases} f(w) & \text{Satisfied} \\ \infty & \text{Otherwise} \end{cases}$

Primal $\min_{w} \theta_p(w)$ | dual

Soln of the primal problem $p^* = \min_{w} \max_{\alpha,\beta} \mathcal{L}(w,\alpha,\beta)$ | $d^* = \max_{\alpha,\beta} \min_{w} \mathcal{L}(w,\alpha,\beta)$

max-min inequality theory

$\max \min f(x) \le \min \max f(x)$

$d^* \le p^*$

$d^* = p^*$

1. f and g shuld be convex and h shuld be affine
2. g_i are (strictly) feasible $g_i() \le 0 \quad \forall i$

there must exist w^*, α^*, β^* and $d^* = p^* = \mathcal{L}(w^*, \alpha^*, \beta^*)$

and w^*, α^*, β^* shuld satisfy KKT

$\frac{\partial \mathcal{L}}{\partial w} = 0 \quad \frac{\partial \mathcal{L}}{\partial \alpha} = 0 \quad \frac{\partial \mathcal{L}}{\partial \beta} = 0 \quad \boxed{\alpha_i^* g_i() = 0}$

$g_i() \le 0$

$\alpha^* \ge 0$

1960's Vladimir Vapnik & Alexey Chervonenkis

1961's $(x,y) \rightarrow$ Kernel $\rightarrow$ transform a non-linear data to a higher dimension with linear separability

$K(x,y) = $

1968 VC dimension : Complexity of the model w.r.t the data point $\Rightarrow$ how can a model be generalize to unknown data.

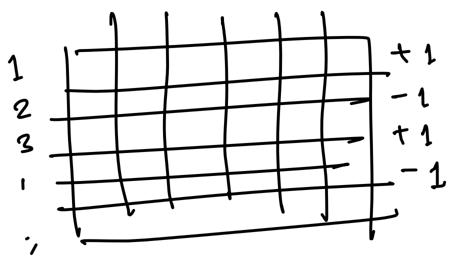
1990 AT&T bell lab

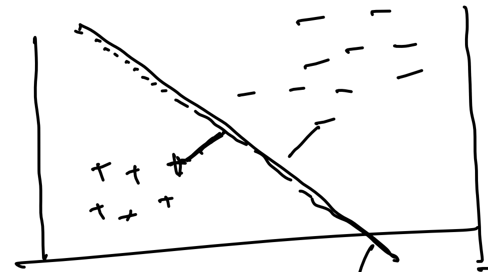
1992 $\boxed{\text{COLT}}$ ← kernel-tricks

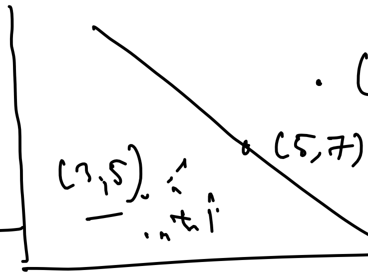
1995 → SVM → Soft-margin

98 – $\boxed{\text{SMO}}$ → Scikit learn

08/09/20







$\leftarrow -0.4x - y + 9 = 0$

$-0.8x - 2y + 18 = 0$

Fun^n margin

$(3,5) \rightarrow \boxed{2.8} \rightarrow 2.8 \times 2$

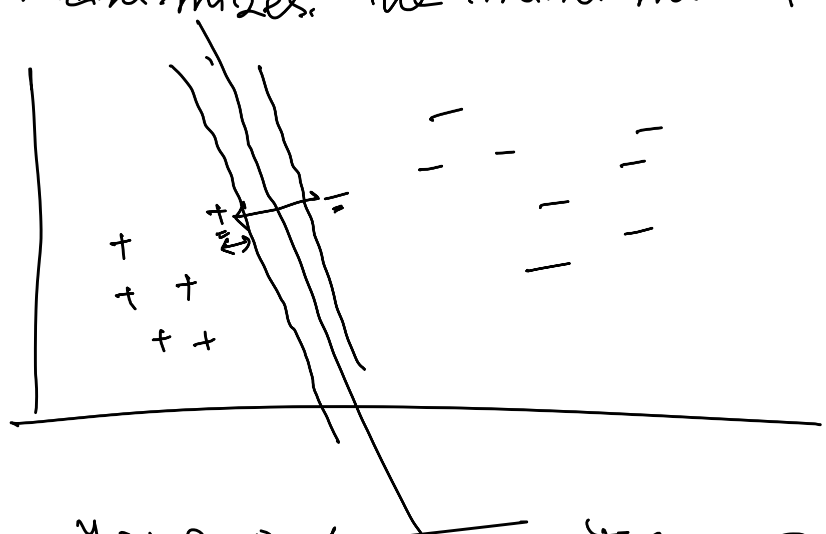
$(5,7) \rightarrow 0$

$(7,9) \rightarrow -2.8 \quad \times 1 \quad 2\uparrow$

Functional Margin:

$$\hat{\gamma}_i = y_i(w^T x_i + b)$$

Objective : To Identify a line that
* maximizes the minimum functional margin.



$-0.4x_i - y_i + 9 = 0 \leftarrow$

$-0.8x_i - 2y_i + 18 = 0 \leftarrow$

Functional margin is "Scale-variant"

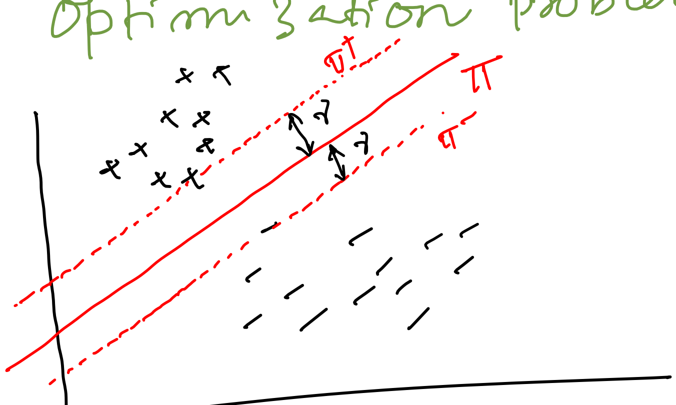
Geometric Margin

$$\gamma = y_i\left(\frac{w^T x_i + b}{\|w\|}\right) \leftarrow \text{Scale invariant}$$

$$= \hat{\gamma}/\|w\|$$

New objective : To identify a line that maximizes the minimum geometric margin.

Optimization problem for Hard margin of SVM



$$\max_{w,b} \gamma \quad \text{s.t.c. geometric Margin}_i \geq \gamma \quad i = 1 \ldots m$$

$$\Rightarrow \max_{w,b} \hat{\gamma}/\|w\|$$

$$\text{s.t.c. } y^i(w^T x_i + b) \geq \hat{\gamma} \quad i = 1 \ldots m$$

$$\max \ 1/\|w\|$$

$$\text{s.t.c. } y^i(w^T x_i + b) \geq 1$$

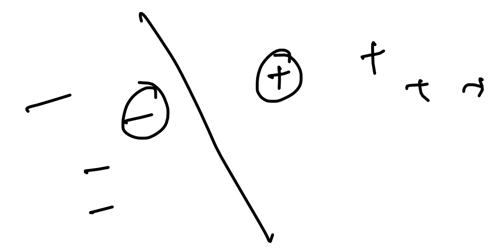
$$\min \frac{1}{2}\|w\|^2 \quad \text{s.t.}$$

$$\boxed{\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{s.t.} \quad y^i(w^T x_i + b) \geq 1}$$

$$\Rightarrow -y^i(w^T x_i + b) + 1 \leq 0 \Leftarrow g_i(w)$$

Instead of solving primal, we will solve dual, and w^*, α^*, β^* ~~sati~~ satisfy KKT

$$\boxed{\alpha_i g_i(w) = 0}$$



(i) $g_i(w) = 0 \Rightarrow$ if $y^i(w^T x_i + b) = 1$

& $\alpha_i > 0$ → SV

(ii) $g_i(w) \neq 0$ $\quad \alpha_i = 0$

$y^i(w^T x_i + b) > 1 \quad g_i(w) \Rightarrow -\text{ve}$

Support vectors: are the data points which are nearest to the decision boundary.

For SV, $\alpha_i > 0$

for other points $\alpha_i = 0$

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{m} \alpha_i \left(y^i(w^T x_i + b) - 1\right) \quad \text{— (1)}$$

Dual problem

$$\max_{\alpha} \min_{w,b} \mathcal{L}(\ )$$

$$\max_{\alpha} \theta_d(\ )$$

$$\theta_d(\alpha) = \min_{w,b} \mathcal{L}(\ )$$

$$\frac{\partial \mathcal{L}}{\partial w} \Rightarrow w - \sum_{i=1}^{m} \alpha_i y^i x_i = 0 \quad \text{— (ii)}$$

$$\frac{\partial \mathcal{L}}{\partial b} \Rightarrow \sum_{i=1}^{m} \alpha_i y^i = 0 \quad \text{— (iii)}$$

Substitute w obtained in (ii) to (1)

$$\frac{1}{2} \sum_{i=1}^{m} \alpha_i y^i x_i \sum_{j=1}^{m} \alpha_j y^j x_j - \left(\sum_{i=1}^{m} \alpha_i \left[y^i \left(\sum_{j=1}^{m} (\alpha_j y^j x_j) x_i + b \right) - 1 \right] \right)$$

$$= \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y^i y^j (x_i \cdot x_j) - \left(\sum_{i=1}^{m} \alpha_i y^i \left(\sum_{j=1}^{m} (\alpha_j y^j x_j) x_i + b \right) \right) + \sum_{i=1}^{m} \alpha_i$$

$$= \sum_{i=1}^{m} \alpha_i + \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y^i y^j (x_i \cdot x_j) - \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y^i y^j (x_i x_j) - \underbrace{b \sum_{i=1}^{m} \alpha_i y^i}_{\to 0}$$

$K_L(x_i, x_j)$

$$= \sum_{i=1}^{m}\alpha_i - \frac{1}{2}\sum_{i=1}^{m}\sum_{j=1}^{m}\alpha_i\alpha_j y^i y^j \underline{\underline{(x_i \cdot x_j)}} \quad — \quad \theta_D$$

$$\boxed{\max_{\alpha} \ \theta_D \quad \text{s.t.} \quad \alpha_i \geq 0, \quad \sum_{i=1}^{m}\alpha_i y^i = 0}$$

$\underline{\underline{K}}(x_i, x_j)$

Kernel

α^*

if α^* is using in (ii) → we will get w^*

From $\alpha_i y_i (w) = 0$ → we will get b^*

Kernel: is a function that takes input as a pair of vectors in the original space and returns dot product of vectors in the transform space.

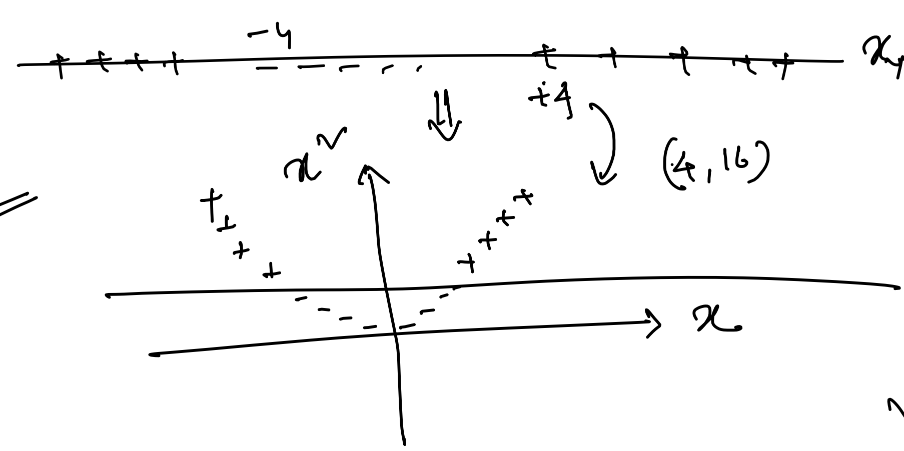
4

$\vec{a}$ $\vec{b}$

(a_1, a_2) (b_1, b_2) ← 2

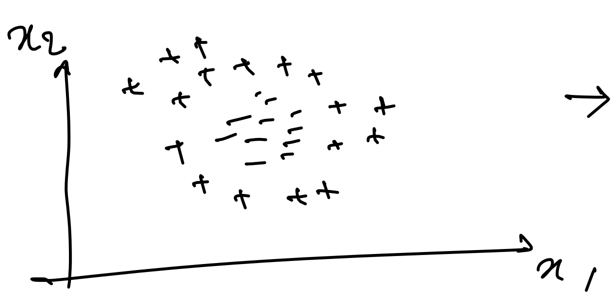
$K(\vec{a}, \vec{b}) \longrightarrow 6$

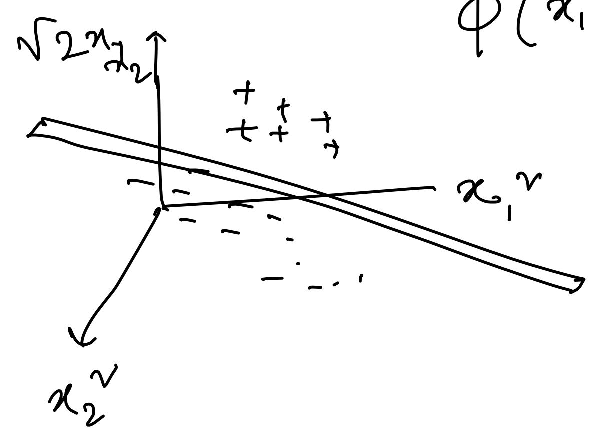
1D



$$\phi(x) = \begin{pmatrix} x \\ x^2 \end{pmatrix}$$

2D





$$\phi(x_1, x_2) = \begin{pmatrix} x_1^2 \\ \sqrt{2}x_1x_2 \\ x_2^2 \end{pmatrix}$$

$a : (a_1, a_2)$

$b : (b_1, b_2)$

$$\phi(a) = \begin{pmatrix} a_1^2 \\ \sqrt{2}a_1a_2 \\ a_2^2 \end{pmatrix}$$

$$\phi(b) = \begin{pmatrix} b_1^2 \\ \sqrt{2}b_1b_2 \\ b_2^2 \end{pmatrix}$$

$$\phi(a) \cdot \phi(b)$$

$$= \begin{pmatrix} a_1^2 \\ \sqrt{2}a_1a_2 \\ a_2^2 \end{pmatrix} \cdot \begin{pmatrix} b_1^2 \\ \sqrt{2}b_1b_2 \\ b_2^2 \end{pmatrix}$$

$$= a_1^2b_1^2 + 2a_1a_2b_1b_2 + a_2^2b_2^2$$

$$= (a_1b_1 + a_2b_2)^2$$

$$= \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}\right)^2 = \underline{\underline{(a \cdot b)^2}}$$

$K_2(a, b)$

Sep 11. 2025

$$\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{s.t.c} \quad y^i(w^T x_i + b) \geq 1 \quad \forall i = 1 \dots m$$

$$\mathcal{L}(\,) = \frac{1}{2}\|w\|^2 - \sum_{i=1}^{m} \alpha_i \left(y^i (w^T x_i + b) - 1 \right)$$

$$\mathcal{L}(\,) = \sum_{i=1}^{m} \alpha_i - \frac{1}{2} \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j y^i y^j \langle x_i \cdot x_j \rangle$$

$$: K(\quad)$$

Linear kernel

$a : (a_1, a_2)$

$b : (b_1, b_2)$

$$K(a, b) = (a \cdot b)^2 = \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \right)^2 \Rightarrow$$

$$\Downarrow$$

$$\phi(a) \cdot \phi(b)$$

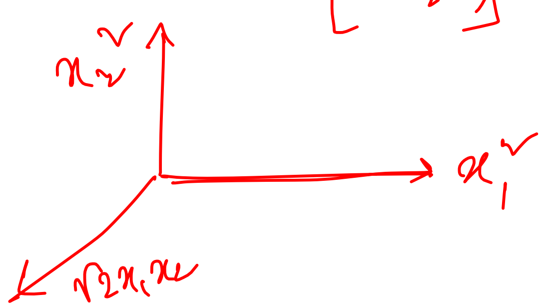
$$\phi(x) = \begin{pmatrix} x_1^2 \\ \sqrt{2} x_1 x_2 \\ x_2^2 \end{pmatrix}$$

$$= \begin{bmatrix} a_1^2 \\ \sqrt{2} a_1 a_2 \\ a_2^2 \end{bmatrix} \cdot \begin{pmatrix} b_1^2 \\ \sqrt{2} b_1 b_2 \\ b_2^2 \end{pmatrix} = a_1^2 b_1^2 + 2 a_1 a_2 b_1 b_2 + a_2^2 b_2^2$$

$$= (a_1 b_1 + a_2 b_2)^2$$

$$= \left(\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \cdot \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \right)^2$$

$x : (x_1, x_2)$



$$K' = (a \cdot b)^3$$

$$\Downarrow$$

$$\phi = \begin{pmatrix} a_1^2 \\ 5\sqrt{a_1 a_2} \\ a_2^2 \end{pmatrix}$$

- Polynomial: $(x_1 x_2 + r)^d \equiv (ab + r)^d$
- * RBF: $\exp(-\gamma \|a - b\|^2)$
- Sigmoid: $\tanh(\gamma\, a \cdot b + c)$
-

Polynomial

$$(a \times b + r)^d$$

r: coeff of the polynomial

d: deg of the polynomial

$r = \frac{1}{2}$

$d = 2$

$$\left(ab + \frac{1}{2}\right)^2 = \left(ab + \frac{1}{2}\right)\left(ab + \frac{1}{2}\right)$$

$$= a^2 b^2 + \frac{1}{2} ab + \frac{1}{2} ab + \frac{1}{4}$$

$$= a^2 b^2 + ab + \frac{1}{4}$$

$$= ab + a^2 b^2 + \frac{1}{4}$$

$$= \begin{pmatrix} a \\ a^2 \\ 1/2 \end{pmatrix} \cdot \begin{pmatrix} b \\ b^2 \\ 1/2 \end{pmatrix} = \begin{pmatrix} a \\ a^2 \end{pmatrix} \cdot \begin{pmatrix} b \\ b^2 \end{pmatrix}$$

$r = 1$
$d = 2$

$$(ab+1)^2 = 2ab + a^2b^2 + 1$$
$$= (\sqrt{2}a, a^2) \cdot (\sqrt{2}b, b^2)$$

$r = 0$
$d = 2$

$$(ab)^2 = a^2 \cdot b^2$$

Keep $r = 0$ and increase d → the dimension will not change but the distance on the 1-D plane will increase.

RBF : Radial Basis Function

$\underline{\underline{e^{-\gamma(a-b)^2}}}$ r: scale of transformation

→ $\gamma = 1$	$a: 2.5$	$b = 4$	$e^{-1(2.5-4)^2} = 0.11$
→ $\gamma = 2$	$a: 2.5$	$b_1 = 4$	→ 0.01
$\gamma = 1$	$a: 2.5$	$b_2 = 16$	$e^{-1(2.5-16)^2} = \cancel{0.00}$ → 0

✓ closure points have higher influence

✓ By varying γ, the influence can be scaled.

$r = 0, d = 2$ $a^2 \cdot b^2$
$+$
$r = 0$ $d = 3$ $a^3 \cdot b^3$
$+$
$r = 0$ $d = 4$ $a^4 \cdot b^4$
$+$
⋮
$r = 0$ $d = \infty$ $a^\infty \cdot b^\infty$

The dimension is not changing!
Is this so?

$$1 + ab + a^2b^2 + a^3b^3 + \dots + a^\infty b^\infty$$
$$= (1, a, a^2, a^3, \dots, a^\infty) \cdot (1, b, b^2, b^3, \dots, b^\infty)$$

Let $\gamma = \frac{1}{2}$ $e^{-\frac{1}{2}(a-b)^2} = e^{-\frac{1}{2}(a^2+b^2-2ab)}$

$$= e^{-\frac{1}{2}(a^2+b^2)} \cdot e^{ab}$$

Taylor series expansion

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^\infty(a)}{\infty!}(x-a)^\infty$$

What is a

a can be any value as long as $f(a)$ exists.

$a = 0$ $f(a) = e^0 = 1$

$$e^x = e^0 + \frac{e^0}{1!}x + \frac{e^0}{2!}x^2 + \dots$$

$$= 1 + \frac{1}{1!}x + \frac{1}{2!}x^2 + \dots$$

$$e^{ab} = 1 + \frac{1}{1!}ab + \frac{1}{2!}(ab)^2 + \frac{1}{3!}(ab)^3 + \dots + \frac{1}{\alpha!}(ab)^\alpha$$

r=0, d=0 r=0, d=1 r=0, d=2 r=0, d=3

$$= \left(1, \sqrt{\tfrac{1}{1!}}\,a, \sqrt{\tfrac{1}{2!}}\,a^2, \dots\right) \cdot \left(1, \sqrt{\tfrac{1}{1!}}\,b, \sqrt{\tfrac{1}{2!}}\,b^2, \dots\right)$$

$$e^{-\frac{1}{2}(a^2+b^2)}\, e^{ab} = \left(s_1 \times 1, s_1\sqrt{\tfrac{1}{1!}}\,a, s_1\sqrt{\tfrac{1}{2!}}\,a^2, \dots\right)\left(s_2 \cdot 1, s_2\sqrt{\tfrac{1}{1!}}\,b, \dots\right)$$

$s_1 = e^{-\frac{1}{2}a^2}$

$s_2 = e^{-\frac{1}{2}b^2}$

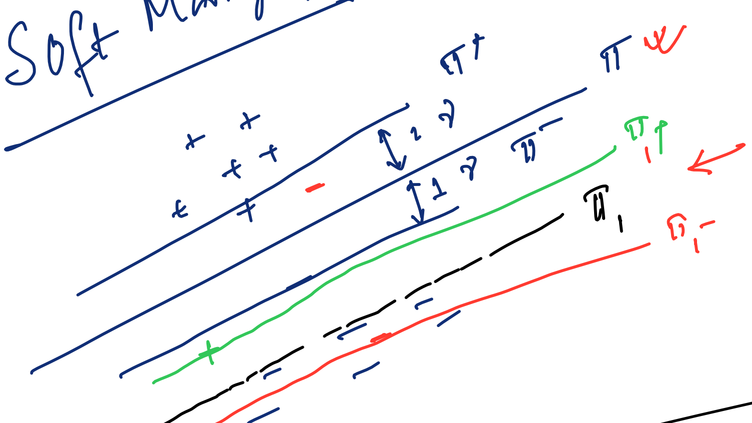
a, b $\quad e^{-\frac{1}{2}(a-b)^2} \rightarrow 0\text{-}1$

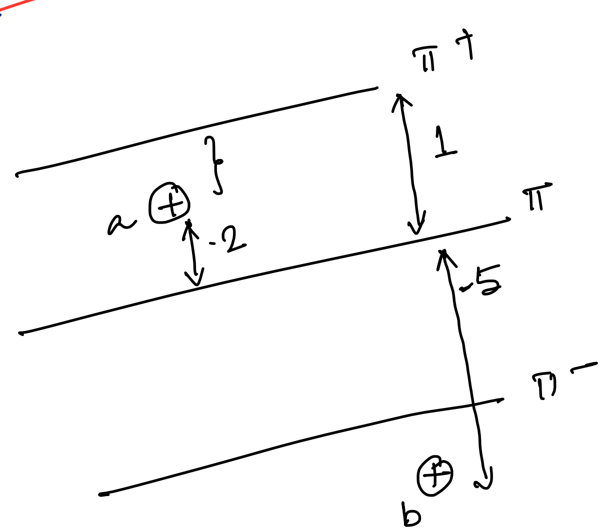
Soft Margin

$$\min \frac{1}{2}\|w\|^2 \quad \text{s.t.c.}$$

$$\min \frac{1}{2}\|w\|^2 + c\,(\text{loss function})^2$$

Constant





$y^i(w^Tx_i + b)$

a: $+1 \times .2 = .2$

loss $1 - .2 = .8$

b: $y^i(w^Tx_i + b)$

−ve

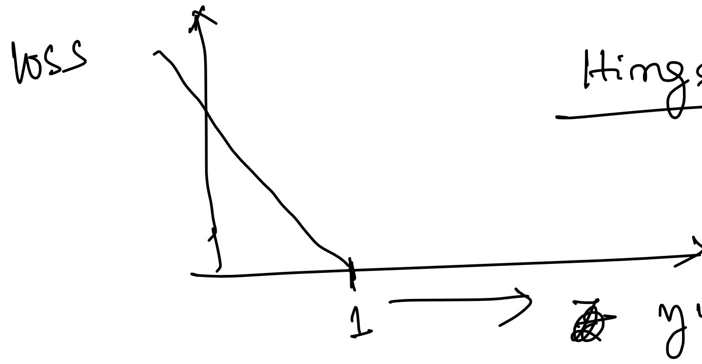
$+1 \times -5 = -5$

loss $1 - (-5) = 6$

$y^i(w^Tx_i + b) \geq 1$ loss = 0

< 1 loss $= 1 - (y^i(w^Tx_i + b))$

$= 1 - Z_i$



Hinge loss

$\rightarrow y^i(w^T x_i + b) \equiv z_i$

$$\text{loss} = \xi_i = \max(0,\ 1 - z_i)$$

$$\min \frac{1}{2}\|w\|^2 + C \sum_{i=1}^{m} \xi_i$$

s.t.c. $y_i(w^T x_i + b) \geq 1 - \xi_i$

Value of ϵ

$C = 10000 \rightarrow$ overfitting

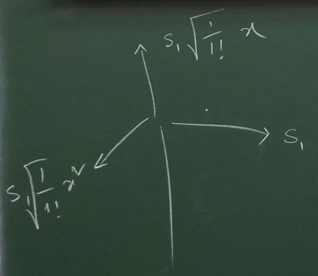
$\frac{1}{2}\|w\|^2 = 200$

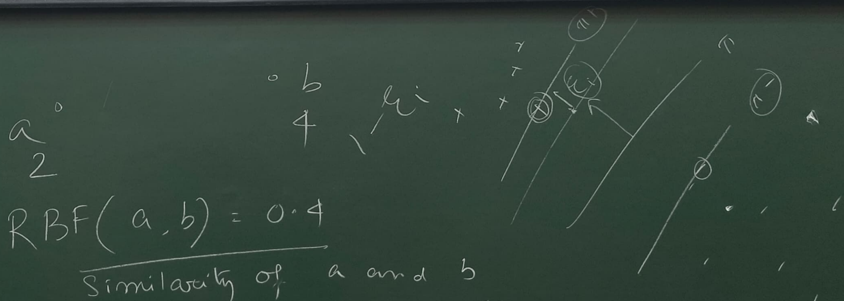
$\sum \xi_i = 2 \leftarrow$ small loss

$O.F = 200 + 10000 \times 2$

$C = 0.001 \rightarrow$ underfit

$\underline{200} + \underline{0.001 \times 2}$





$$RBF(a, b) = 0.4$$

Similarity of a and b
in the infinite dimensional space.
when the dimensions are